Previous University questions & Answers

Subject: EE 309 Microprocessor and Embedded system

Module 2

December 2017

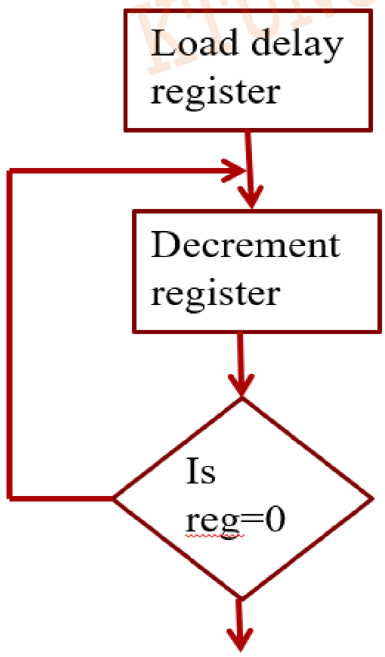
1. Write a delay subroutine program in 8085 for 0.4 ms. Assume the clock frequency as 3 MHz. (5)

   Consider the clock frequency 3MHZ

   Clock period, $T = 1/f = 1/(3*10^6) = 0.33\mu s$

   We can generate time delay by using following program:

```
        MVI C,57H
LOOP    DCR C
        JNZ LOOP
```

We need to calculate the count value

Time delay in loop: TL= (T * T-states * N10)

Where; T= clock period = T = 1/f = 1/(3*10^6) = 0.33μs

T-state= Total no. of T-state within loop

Label	Mnemonics	T-state
	MVI C,57H	7
LOOP	DCR C	4
	JNZ LOOP	10/7

Total number of T-state within loop = 14

Here time delay given as 4ms. So, we need to calculate count value for corresponding delay

$$0.4\,ms = 0.33 \times 10^{-6} \times 14 \times N_{10}$$

$$N_{10} = \frac{0.4 \times 10^{-3}}{0.33 \times 10^{-6} \times 14}$$

$$= \frac{0.4 \times 10^{3}}{0.33 \times 14} = \frac{400}{462} = 86.5 = \underline{\underline{87}}$$

$\boxed{N_{10} = 87}$ → corresponding hex value = 57

$\boxed{\text{so count value} = 57\,H}$

By using 57 H as count value , we can generate 0.4 ms time delay.

2. Draw and explain the timing diagram of LDAX B. (10)

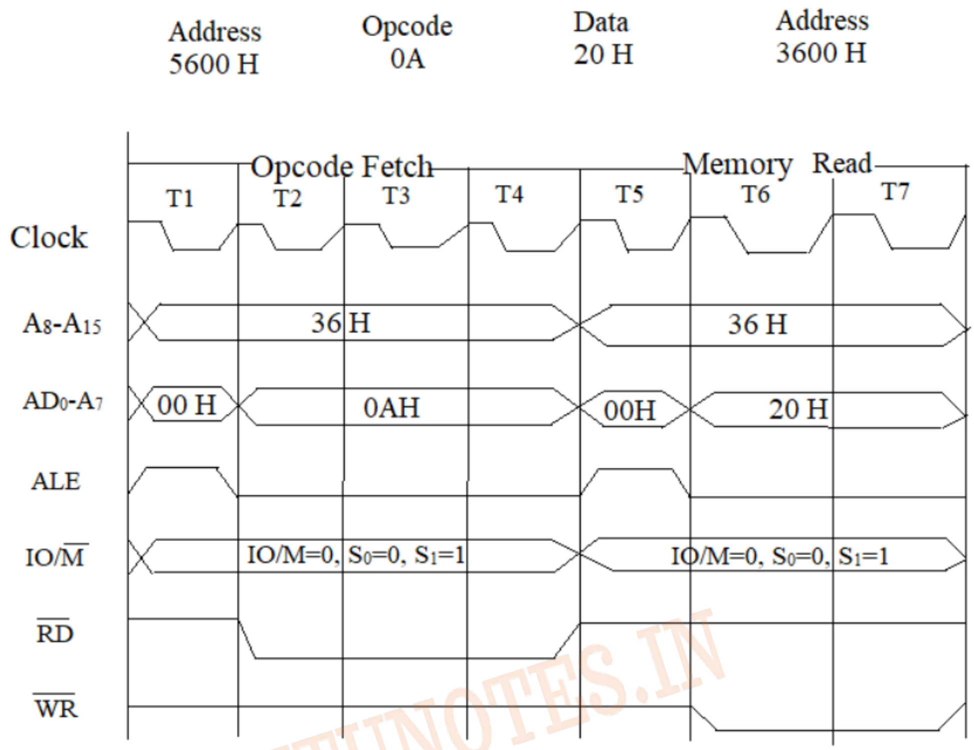


Fig: Timing Diagram of LDAX B

Instruction: LDAX B

Opcode: 0AH

Opcode stored in address location :5600

LDAX B: load content of memory location pointed by B register pair

For example: If content of B -3600, then data stored in memory location 3600 will load to A register.

Timing diagram consists of two machine cycle.

Machine cycle 1: **opcode fetch** – opcode 0AH fetched from stored memory location, says 5600H. 4 T-state needed.

During T1, lower byte of memory address, 00H, placed on AD0-AD7 & higher order byte of address, 56H, placed on A8-A15. ALE become high, since address placed on address/data bus.

During T2 & T3, opcode, 0AH, placed on address/data bus. ALE becomes low. Read signal become low, since opcode read from memory.

During T4, opcode decoding occurs.

Machine cycle 2: Memory read

Content of memory location 3600, which is pointed by B register pair, loaded to A register.

During T1, address placed on AD0-AD7 & A8-A15.

During T2 & T3, data (20H) read from memory location 3600H

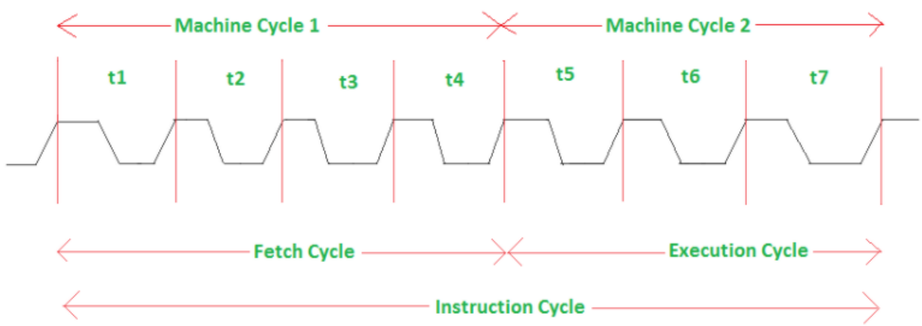
IO/M	S0	S1	Machine cycle
0	1	1	Opcode fetch
0	0	1	Memory read
0	1	0	Memory write

3. Explain the terms Machine cycle and T-states. (4)

Answer:

Machine cycle: The time required by the microprocessor to complete an operation of accessing memory or input/output devices is called *machine cycle*.

T-state: One time period of frequency of microprocessor is called *t-state*. A t-state is measured from the falling edge of one clock pulse to the falling edge of the next clock pulse. Fetch cycle takes four t-states and execution cycle takes three t-states.

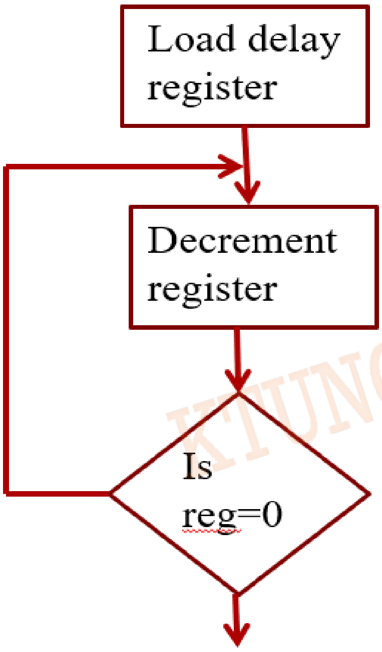


December 2018

4. Explain how delay is generated using one register and calculate how much delay can be generated if one register is loaded with maximum count. Assume time for one state is 320ns. (5)

Time delay using one register

- A count is loaded in a register
- Loop is executed until the count reaches to zero.



Label	Mnemonics	T-state
	MVI C,FFH	7
LOOP	DCR C	4
	JNZ LOOP	10/7

Here clock period given as 320ns. If we store maximum value (FF) in count register , can generate delay:

Time delay in LOOP

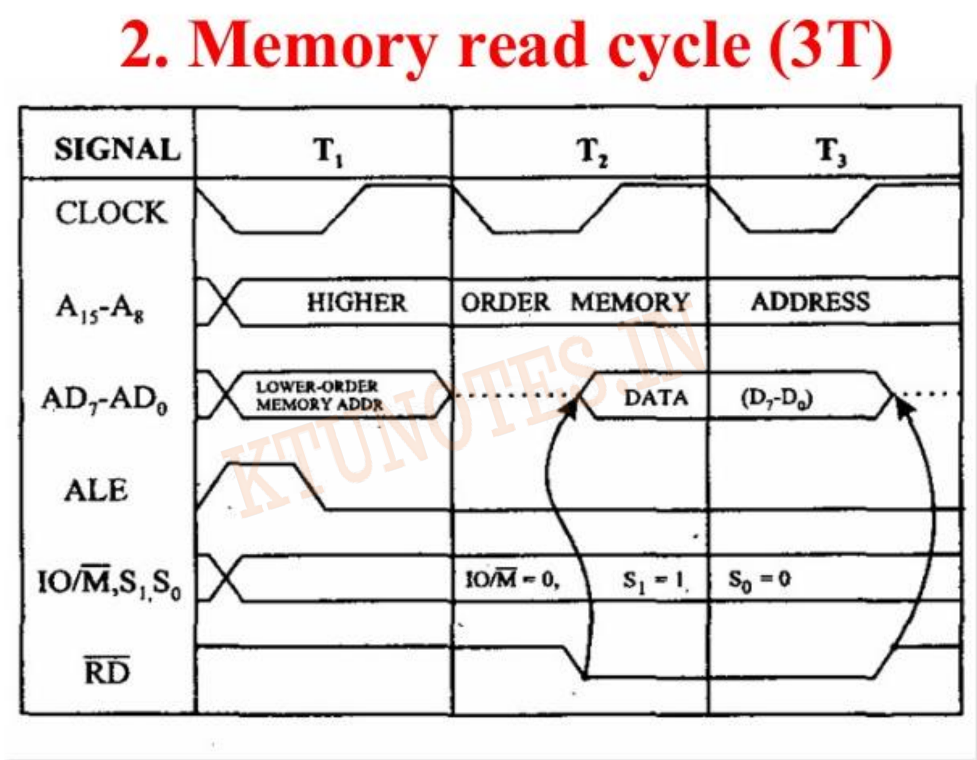
TL= (T * T-states * N10)

$= 320*10^{-9}* 14*255$

= 1142400 ns

= 1.14 ms

5. **Draw the timing diagram for Memory Read operation. (5)**



Memory read machine cycle need 3 T-states.

A15-A8 – During T1, T2 & T3 , higher order memory address place.

AD7-AD0 – During T1, lower order memory address place. During T2 & T3 data placed.

ALE- high during T1

IO/M – zero, since the current operation is memory read

S1,S0- status signal. S0S1= 01 during read operation.

RD – read signal low during read operation.

6. **Explain CALL & RETURN instructions in 8085. (5)**

Two instructions to implement subroutine are:

i. CALL
ii. RETURN

CALL:

- Used in the main program to call a subroutine.
- 3- byte instruction that transfer program sequence to a subroutine address
- Save content of program counter into stack
- Jump unconditionally to the specified memory location

RET:

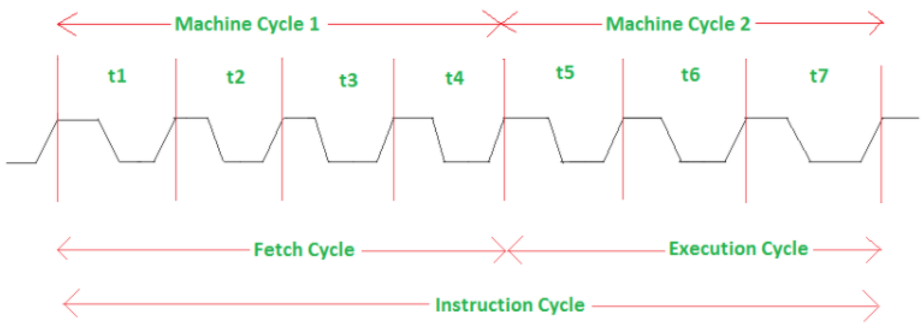
- 1 byte instruction
- Inserted top of stack into program counter
- Unconditionally returns from subroutine

Main program	2000	LXI SP, 2400 H	Initialize stack pointer
	2040	CALL 2070H	Call subroutine located at 2070H
	2041		
	2042		
	2043	Next Instruction	Address of next instruction
	205F	HLT	End of man program
sub program	2070	Subroutine	Beginning of subroutine
	207F	RET	End of subroutine
stack	23FF		
	2400		Stack pointer location

7. **Explain Machine Cycle and Instruction Cycle (5)**

Machine cycle: The time required by the microprocessor to complete an operation of accessing memory or input/output devices is called *machine cycle*.

T-state: One time period of frequency of microprocessor is called *t-state*. A t-state is measured from the falling edge of one clock pulse to the falling edge of the next clock pulse. Fetch cycle takes four t-states and execution cycle takes three t-states.



Instruction cycle: Time required to execute and fetch an entire instruction is called *instruction cycle.* It consists:

- **Fetch cycle –** The next instruction is fetched by the address stored in program counter (PC) and then stored in the instruction register.
- **Decode instruction –** Decoder interprets the encoded instruction from instruction register.
- **Reading effective address –** The address given in instruction is read from main memory and required data is fetched. The effective address depends on direct addressing mode or indirect addressing mode.
- **Execution cycle –** consists memory read (MR), memory write (MW), input output read (IOR) and input output write (IOW)

December 2019

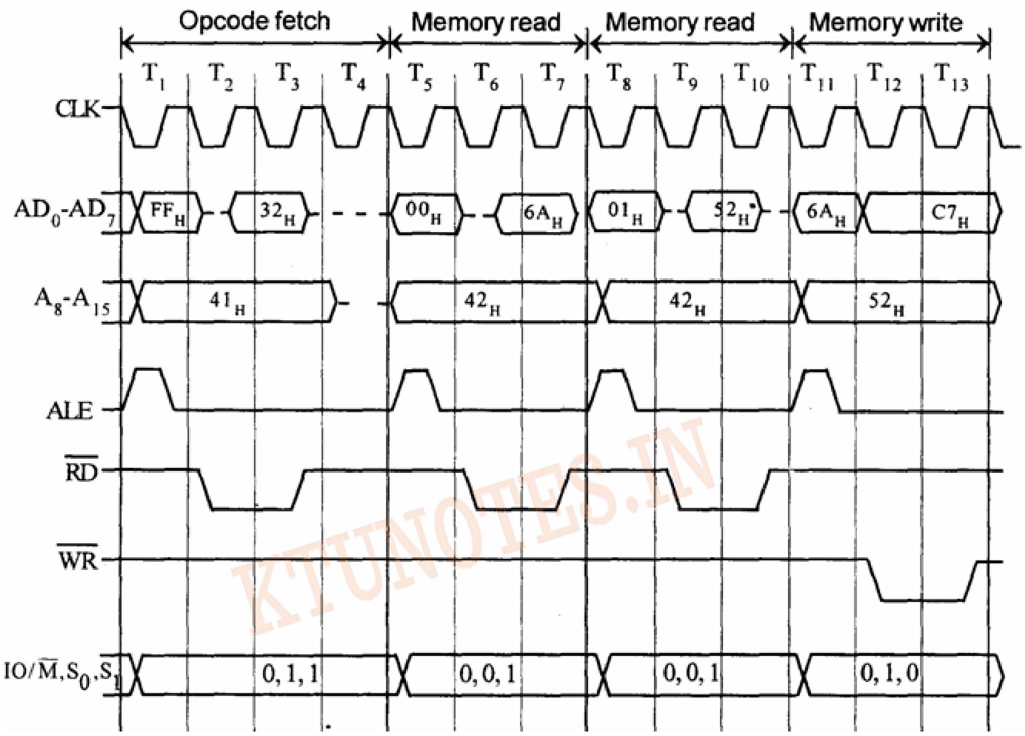
8. Explain subroutine CALL and RET instructions in 8085 (5)-Repeated question

9. Draw the timing diagram of instruction STA 4500H. (10)

Timing diagram for STA 526AH.

- STA means Store Accumulator -The contents of the accumulator is stored in the specified address(526A).
- The opcode of the STA instruction is said to be 32H. It is fetched from the memory location; for e.g. 41FFH – **1st machine cycle opcode fetch**
- Then the lower order memory address is read(6A). - **Memory Read Machine Cycle**
- Read the higher order memory address (52).- **Memory Read Machine Cycle**

- The combination of both the addresses are considered and the content from accumulator is written in 526A. - **Memory Write Machine Cycle**
- Assume the content of accumulator is C7H. So, C7H from accumulator is now stored in 526A H.



Address	Mnemonics	Opcode
41FF	STA 526AH	32H
4200		6AH
4201		52H

Timing diagram consists of four machine cycle.

Machine cycle 1: **opcode fetch** – opcode 32H fetched from stored memory location 41FF

During T1, lower byte of memory address, FFH, placed on AD0-AD7 & higher order byte of address, 41H, placed on A8-A15. ALE become high, since address placed on address/data bus.

During T2 & T3, opcode placed on address/data bus. ALE becomes low. Read signal become low, since opcode read from memory.

During T4, opcode decoding occurs.

Machine cycle 2: Memory read

During T1, lower byte of memory address, 00H, placed on AD0-AD7 & higher order byte of address, 41H, placed on A8-A15. ALE become high, since address placed on address/data bus.

During T2 & T3: lower byte of address 6A placed on AD0-AD7

Machine cycle 3: Memory read

During T1, lower byte of memory address, 01H, placed on AD0-AD7 & higher order byte of address, 41H, placed on A8-A15. ALE become high, since address placed on address/data bus.

During T2 & T3: lower byte of address 52 placed on AD0-AD7

Machine cycle 3: Memory write: accumulator content to memory address 526A H

During T1, lower byte of memory address, 6AH, placed on AD0-AD7 & higher order byte of address, 52H, placed on A8-A15. ALE become high, since address placed on address/data bus.

During T2 & T3: Accumulator content stored to memory

IO/M	S0	S1	Machine cycle
0	1	1	Opcode fetch
0	0	1	Memory read
0	1	0	Memory write

10. Explain Fetch cycle & Execute cycle in 8085. (4)-Repeated question

April 2018

11. Explain the PUSH and POP instructions of 8085 with example. (5)

- Stack is a group of memory locations in RAM
- Used for temporary storage of data during the execution of program.
- Starting location of stack is defined in the main program, using instruction LXI SP, 16 bit

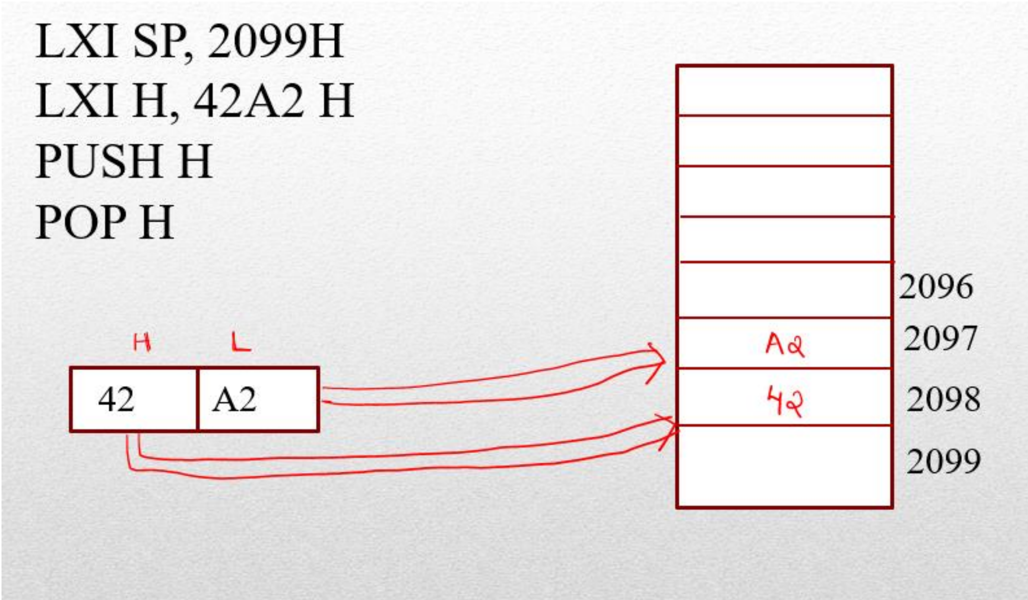
- Storing of data bytes begins at the memory location , that is one less than address stored in stack pointer
- E.g. LXI SP, 2045H ; Then storing of data byte begins at 2043H
- Continues to store in reverse order

PUSH & POP :

- Data byte in register pair can be stored in stack using instruction PUSH Rp
  E.g. PUSH B
- Data bytes can be transferred from stack to register by using instruction POP Rp
  E.g. POP B
  Consider the program:
  LXI SP, 2099H
  LXI H, 42A2 H
  PUSH H
  POP H

When PUSH executed:

- Stack pointer decremented by one to 2098H
- Content of H register copied in to 2098H
- SP again decremented by one to 2097H
- Content of L register copied in to 2097H



When POP executed:

- Content of top of stack (location shown by SP) are copied into L register
- Stack pointer incremented by one to 2098H
- Content of 2098H copied in to H register
- Stack pointer incremented by one to 2099H

12. Define instruction cycle and machine cycle. (4)-Repeated question

13. Draw and explain the timing diagram for the instruction MOV C, A (opcode: 4FH), stored in location 6000H.

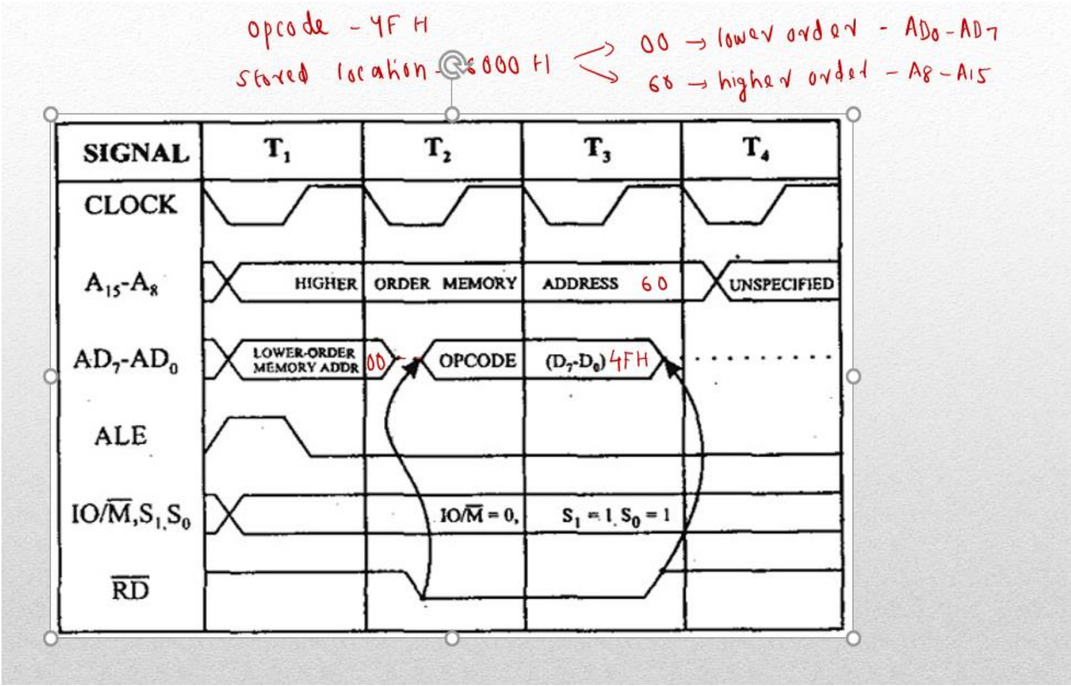
Instruction: MOV C, A

Opcode: 4FH

Opcode stored in address location :6000H

MOV C, A: Move accumulator content to C register

One machine cycle needed to complete execution: Opcode fetch



Timing diagram consists of one machine cycle.

Machine cycle 1: **opcode fetch** – opcode 4FH fetched from stored memory location 6000H

During T1, lower byte of memory address, 00H, placed on AD0-AD7 & higher order byte of address, 60H, placed on A8-A15. ALE become high, since address placed on address/data bus.

During T2 & T3, opcode, 4FH, placed on address/data bus. ALE becomes low. Read signal become low, since opcode read from memory.

During T4, opcode decoding occurs& content of A register moved to C register.

IO/M	S0	S1	Machine cycle
0	1	1	Opcode fetch
0	0	1	Memory read
0	1	0	Memory write